

Answers to Theory Assignment 1

1. [20 points] Fill in all the missing values. For column A you need to compute the sums. For column B (the last two rows) you need to guess a function that does not contradict any of the yes/no answers already in the next three columns. Fill in each empty entry in the last three columns with either a yes/no answer.

Function	Function	O	Ω	Θ
A	B	$A = O(B)$	$A = \Omega(B)$	$A = \Theta(B)$
n^4	$n^3 \lg n$	no	yes	no
$n\sqrt{n}$	n^2	yes	no	no
$(n+1)!$	$n!$	no	yes	no
$\lg n$	n^k where $k > 0$	yes	no	no
$\sum_{i=1}^n (i+1) = \frac{n(n+1)}{2} + n$	n^2	yes	yes	yes
$\sum_{i=0}^{n-1} 3^i = \frac{3^n - 1}{3 - 1}$	n^4 (or any polynomial)	no	yes	no

2. [20 points] Prove the following using the original definitions of O, Ω , θ , o, and ω .
- $9n^3 + 67n^2 + 8n - 10 \in O(n^3)$
 Answer: For $n \geq 1$, $9n^3 + 67n^2 + 8n - 10 \leq 9n^3 + 67n^3 + 8n^3 = 84n^3$. Thus, $9n^3 + 67n^2 + 8n - 10 \in O(n^3)$
 - $4087n^3 \in \Omega(n^2)$
 Answer: For $n \geq 1$, $4087n^3 \geq n^2$. Thus, $4087n^3 \in \Omega(n^2)$
 - $14n^3 + 5n^2 \in \omega(n^2)$
 To prove this, we need to show that $14n^3 + 5n^2 \geq c n^2$ for any $c \geq 0$. If we divide both sides of the inequality by n^2 , we get $14n + 5 \geq c$, i.e., $n \geq (c-5)/14$. Thus, pick any $N \geq \max((c-5)/14, 1)$ that conclude the proof. (For any c, you can pick an N value that satisfies the inequality.)
 - $46n^3 \in o(n^4)$
 To prove this, we need to show that $46n^3 \leq c n^4$ for any $c \geq 0$. If we divide both sides of the inequality by n^3 , we get $46 \leq cn$, i.e. $46/c \leq n$. Thus, pick any $N \geq 46/c$ to conclude the proof. (For any c, you can pick an N value that satisfies the inequality.)
 - $n^2 + 3n - 10 \in \Theta(n^2)$
 (1) First, $n^2 + 3n - 10 \leq n^2 + 3n \leq 4n^2$ for all $n \geq 1$, that is $n^2 + 3n - 10 \leq cn^2$ for constants $c = 4$ and $n_0 = 1$ and $n \geq n_0$. Therefore, $n^2 + 3n - 10 \in O(n^2)$

- (2) Next, $n^2 + 3n - 10 = n^2 + (3n - 10) \geq n^2$ when $3n - 10 \geq 0$, which holds when $n \geq 4$. That is $n^2 + 3n - 10 \geq cn^2$ for constants $c = 1$ and $n_0 = 4$ and $n \geq n_0$. Therefore, $n^2 + 3n - 10 \in \Omega(n^2)$.

In summary, for $c_1 = 1$, $c_2 = 4$ and $n_0 = 4$, we have $c_1 n^2 \leq n^2 + 3n - 10 \leq c_2 n^2$ when $n \geq n_0$. Therefore, $n^2 + 3n - 10 \in \Theta(n^2)$ according to the definition of Θ .

3. [15 points] Prove the following using limits.

- (a) $n^{1/n} \in \Theta(1)$ [Hint: you can use $x = e^{\ln x}$]

$$\text{Answer: } n^{1/n} = e^{1/n \ln(n)}. \lim_{n \rightarrow \infty} \frac{1 \ln(n)}{n} = 0. \lim_{n \rightarrow \infty} e^{\frac{\ln n}{n}} = 1$$

Or you can use the following:

Let $x = n^{1/n}$. Then $\lg x = (1/n) \lg n = (\lg n) / n$. We know $\lim_{n \rightarrow \infty} \frac{\lg n}{n} = 0$. This means that when n becomes infinity, $\lg x = 0$, which in turn implies that $x = 1$ when n becomes infinity, $\rightarrow$

$$\lim_{n \rightarrow \infty} n^{1/n} = 1.$$

$$\begin{aligned} \text{(b) } 4^n &\in \omega(n^k) \\ &= \lim_{n \rightarrow \infty} \frac{4^n}{n^k} = \lim_{n \rightarrow \infty} \frac{4^n \ln 4}{k n^{k-1}} \text{ using L'Hopitals rule} \\ &= \lim_{n \rightarrow \infty} \frac{4^n \ln 4}{k(k-1) n^{k-2}} \text{ using L'Hopitals rule} \\ &= \lim_{n \rightarrow \infty} \frac{4^n (\ln 4)^k}{k!} \\ &= \infty \end{aligned}$$

- (c) $\lg^3 n \in o(n^{0.5})$

Answer: $\lim[\lg^3 n / n^{0.5}] = \lim[(3 \lg^2 n / (n \ln 2)) / (0.5 / n^{0.5})] = C_1 * \lim[\lg^2 n / n^{0.5}] = \dots = C_2 \lim[\lg n / n^{0.5}] = \dots = C_3 \lim[1 / n^{0.5}] = 0$, and from Theorem 1.3 this implies that $\lg^3 n \in o(n^{0.5})$.

4. [10 points] Order the functions below by increasing growth rates (no justification required):

$$n^n, n, n \ln n, n^{1/2}, 2^{\lg n}, \ln n, 18, n^{1/n}, \sqrt{2}^{\lg n}, n!, \lg(n^{10}), 2^n$$

Let $g_i(n)$ be the i th function from the left after the ordering (the leftmost function has the slowest growth rate). In the order, $g_i(n)$ should satisfy $g_i(n) \in O(g_{i+1}(n))$. If two or more functions are equivalent (in terms of Θ), put them in [] separated by comma (e.g., $[n^2, 5n^2]$).

Answer: The order is: $[18, n^{1/n}]$, $[\ln n, \lg(n^{10})]$, $[n^{1/2}, \sqrt{2}^{\lg n}]$, $[n, 2^{\lg n} (= n^{\lg 2} = n)]$, $n \ln n$, 2^n , $n!$, n^n

$$\sqrt{2}^{\lg n} = 2^{(1/2) \lg n} = 2^{\lg \sqrt{n}} = \sqrt{n}$$

Both 10 and $n^{1/n}$ are in $\Theta(1)$. It can be shown that $\lim_{n \rightarrow \infty} n^{1/n} = 1$. Let $x = n^{1/n}$. Then $\lg x = (1/n)\lg n = (\lg n)/n$. We know $\lim_{n \rightarrow \infty} \frac{\lg n}{n} = 0$. This means that when n becomes infinity, $\lg x = 0$, which in turn implies that $x = 1$ when n becomes infinity, $\Rightarrow \lim_{n \rightarrow \infty} n^{1/n} = 1$.

That $n! \in O(n^n)$ can be shown using Stirling's Approximation. From this approximation, we know $n! \leq 2\sqrt{2\pi n} \left(\frac{n}{e}\right)^n$. Thus $\lim_{n \rightarrow \infty} \frac{n!}{n^n} \leq \lim_{n \rightarrow \infty} \frac{2\sqrt{2\pi n} \left(\frac{n}{e}\right)^n}{n^n} = \lim_{n \rightarrow \infty} \frac{2\sqrt{2\pi n}}{e^n} = 0$. Therefore, $n! \in O(n^n)$.

The other relationships are either easy to see or have been discussed in class.

5. [20 points] Let $f(n)$ and $g(n)$ be asymptotically positive functions. For each of the following conjectures, either prove it is true or provide a counter example to show it is not true.
- $(f(n) + g(n)) \in \Theta(\max(f(n), g(n)))$.
 - $f(n) \in O(g(n))$ implies $2^{f(n)} \in O(2^{g(n)})$.
 - $f(n) \in O(g(n))$ implies $g(n) \in \Omega(f(n))$.
 - $f(n) \in O(g(n))$ implies $\lg(f(n)) \in O(\lg(g(n)))$, where $\lg(g(n)) \geq 1$ and $f(n) \geq 1$ for sufficiently large n .

Answer: a) is true. Proof: It is easy to see that $\max(f(n), g(n)) \leq (f(n) + g(n)) \leq 2 \cdot \max(f(n), g(n))$ for any n . Based on the definition of Θ , $(f(n) + g(n)) \in \Theta(\max(f(n), g(n)))$ is true.

b) is not true. A counter example is as follows. Let $f(n) = 2n$ and $g(n) = n$. Then $f(n) \in O(g(n))$ is true. However, $2^{f(n)} = 2^{2n} = 4^n$ grows faster than $2^{g(n)} = 2^n$. So $2^{f(n)} \in O(2^{g(n)})$ is not true.

c) is true. Proof: If $f(n) \in O(g(n))$, then there exist positive constants c and n_0 , such that $f(n) \leq cg(n)$ for all $n \geq n_0$. This means that there is a positive constant $c_1 = 1/c$ and constant n_0 which make $g(n) \geq (1/c)f(n) = c_1 f(n)$ for $n \geq n_0$, that is, $g(n) \in \Omega(f(n))$ is true.

d) is true. Proof: If $f(n) \in O(g(n))$, then there exist positive constants c and n_0 , such that $f(n) \leq cg(n)$ for all $n \geq n_0$. $\lg(f(n)) \leq \lg(c) + \lg(g(n)) \leq \lg(c) + \lg(g(n)) \leq (\lg(c) + 1)\lg(g(n))$

6. [10 points] Prove that for all integers $n > 0$,

$$\left(\sum_{i=1}^n i \right)^2 = \sum_{i=1}^n i^3.$$

by mathematical induction. Divide your proof into the three required parts: Induction Base, Induction Hypothesis, and Induction Steps.

Answer:

Induction base: For $n=1$, $(1)^2=1^3$.

Induction hypothesis: Assume for arbitrary positive integer n , that

$$\left(\sum_{i=1}^n i\right)^2 = \sum_{i=1}^n i^3.$$

Induction step: We need to show

$$\left(\sum_{i=1}^{n+1} i\right)^2 = \sum_{i=1}^{n+1} i^3$$

To that end,

$$\begin{aligned}\left(\sum_{i=1}^{n+1} i\right)^2 &= \left(\left(\sum_{i=1}^n i\right) + (n+1)\right)^2 = \left(\sum_{i=1}^n i\right)^2 + (n+1)^2 + 2(n+1) \sum_{i=1}^n i \\&= \left(\sum_{i=1}^n i\right)^2 + (n+1)^2 + 2(n+1) \frac{n(n+1)}{2} = \left(\sum_{i=1}^n i\right)^2 + (n+1)^2 + (n+1)^2 n \\&= \left(\sum_{i=1}^n i\right)^2 + (n+1)^3 = \sum_{i=1}^n i^3 + (n+1)^3 = \sum_{i=1}^{n+1} i^3\end{aligned}$$

7. [15 points] Consider the following algorithm:

```

for ( i = 2; i <= n; i++) {
    for ( j = 0; j <= n) {
        cout << i << j;
        j = j + ⌊n/4⌋;
    }
}

```

(a) What is the output when $n=4$?

(b) What is the time complexity $T(n)$. You may assume that n is divisible by 4.

Answer:

(a) $n = 4$: 2 0 2 1 2 2 2 3 2 4 3 0 3 1 3 2 3 3 3 4 4 0 4 1 4 2 4 3 4 4

(b) $T(n) = (n-1)(5) \in O(n)$ because the inner loop will always have 5 loops.

8. [10 points] What is the time complexity $T(n)$ of the nested loops below? For simplicity, you may assume that n is a power of 2. That is, $n = 2^k$ for some positive integer k . Give some justification for your answer.

```

for ( i = 1; i <= n; i++){
    j = n;
    while ( j >= 1){
        < body of the while loop >    //Needs  $\Theta(1)$ .
        j = ⌊j/2⌋;
    }
}

```

Answer:

$T(n) = n \cdot \log n = O(n \cdot \log n)$